

Md Ashiqur Rahman

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Research Interest

Designing robust and physically grounded AI systems, spanning geometric and scale-equivariant deep learning, generative models and neural operators for scientific computing, and vision-language models for robust perception and reasoning.

Education

Purdue University **West Lafayette, IN**
Doctor of Philosophy in Computer Science 2021–

Advisor: Raymond A. Yeh

[Awarded Bilsland Dissertation Fellowship \(2026–2027\)](#)

Highly competitive, nomination-based fellowship for top final-year PhD candidates (1–2 per department).

Purdue University **West Lafayette, IN**
Master of Science in Computer Science 2021–2025

Bangladesh University of Engg & Tech **Dhaka, Bangladesh**
Bachelor of Science in Computer Science and Engineering 2015–2019

Advisor: Md. Shamsuzzoha Bayzid

Professional Employment

Purdue University **West Lafayette, IN**
Graduate Assistant in the Department of Computer Science 2021–present

Eli Lilly **Indianapolis, IN**
Research Intern Summer 2026

Advanced Intelligence Team

- Developed an edge-deployed agentic-vision system for monitoring pharmaceutical manufacturing.

Autodesk **San Francisco, CA**
Research Intern Summer 2025

Manager: Robert Giaquint

- Investigated the reliability of VLMs for 3D CAD model generation for engineering.

NVIDIA **Santa Clara, CA**
Research Intern Summer & Fall 2023

Manager: Anima Anandkumar

- Worked on large-scale weather prediction and foundation models for scientific computing.

United International University **Dhaka Bangladesh**
Lecturer in the Department of Computer Science 2019–2021

Publications

Conference

- [C1] **Md Ashiqur Rahman**, Jun Hao Lim, Jeremiah Jiang, Teck-Yian Lim, and Raymond A. Yeh. Tunable soft equivariance with guarantees. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE/CVF, 2026.
- [C2] **Md Ashiqur Rahman**, Chiao-An Yang, Michael N. Cheng, Jun Hao Lim, Jeremiah Jiang, Teck-Yian Lim, and Raymond A. Yeh. Local scale equivariance with latent deep equilibrium canonicalizer. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2025.
- [C3] Tinghan Yang, **Md Ashiqur Rahman**, and Raymond A. Yeh. Clipsym: Delving into symmetry detection with clip. In *International Conference on Computer Vision (ICCV)*, 2025.
- [C4] **Md Ashiqur Rahman** and Raymond A. Yeh. Group downsampling with equivariant anti-aliasing. In *International Conference on Learning Representations (ICLR)*, 2025.

- [C5] **Md Ashiqur Rahman**, R. J. George, M. Elleithy, D. Leibovici, Z. Li, B. Bonev, C. White, J. Berner, R. A. Yeh, J. Kossai, K. Azizzadenesheli, and A. Anandkumar. Pretraining codomain attention neural operators for solving multiphysics PDEs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2024.
- [C6] **Md Ashiqur Rahman** and Raymond A. Yeh. Truly scale-equivariant deep nets with fourier layers. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2023.
- [C7] **Md Ashiqur Rahman**, Abdullah Aman Tutul, and A. B. M. Alim Al Islam. Solving the maze of diagnosing parkinson’s disease based on portable eeg sensing to be adaptable to go in-the-wild. *Proceedings of the 7th International Conference on Networking, Systems and Security*, 2020. (Best Paper Award).
- [C8] Akm Ashiquzzaman, Abdul Kawsar Tushar, **Md Ashiqur Rahman**, and Farzana Mohsin. An efficient recognition method for handwritten arabic numerals using cnn with data augmentation and dropout. *Data Management, Analytics and Innovation*, 2018.

Journal

- [J1] Hrishikesh Viswanath, **Md Ashiqur Rahman**, Abhijeet Vyas, Andrey Shor, Beatriz Medeiros, Stephanie Hernandez, Suhas Eswarappa Pameela, and Aniket Bera. Neural operator: Is data all you need to model the world? an insight into the paradigm of data-driven scientific ml. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, pages 1–18, 2026.
- [J2] **Md Ashiqur Rahman**, Abdullah Aman Tutul, Mahfuza Sharmin, and Md. Shamsuzzoha Bayzid. Beene: deep learning-based nonlinear embedding improves batch effect estimation. *Bioinformatics*, 2023.
- [J3] **Md Ashiqur Rahman**, Manuel A. Florez, Anima Anandkumar, Zachary E. Ross, and Kamyar Azizzadenesheli. Generative adversarial neural operators. *Transactions on Machine Learning Research*, 2022.
- [J4] **Md Ashiqur Rahman**, Zachary E. Ross, and Kamyar Azizzadenesheli. U-no: U-shaped neural operators. *Transactions on Machine Learning Research*, 2022.
- [J5] **Md Ashiqur Rahman**, Abdullah Aman Tutul, Sifat Muhammad Abdullah, and Md. Shamsuzzoha Bayzid. Chapao: Likelihood and hierarchical reference-based representation of biomolecular sequences and applications to compressing multiple sequence alignments. *PLoS ONE*, 2022.

Under Review

- [S1] Kaivalya Agrawal, **Md Ashiqur Rahman**, Raymond A. Yeh, and Zachary Kingston. Skipvla: Skipping vla steps with classical planning for fast robot manipulation. Under Review.
- [S2] Hrishikesh Viswanath, **Md Ashiqur Rahman**, Chi Lin, Damon Conover, and Aniket Bera. Hessianforge: Scalable lidar reconstruction with physics-informed neural representation and smoothness energy constraints. Under Review.

Patents

- [P1] Aniket Bera, Rashmi Bhaskara, **Md Ashiqur Rahman**, and Hrishikesh Viswanath. Physics-informed adaptive fourier neural interpolation operator for synthetic frame generation, January 9 2025. US Patent App. 18/758,927.

Awards & Recognition

- Bilsland Dissertation Fellowship** 2026–2027
 - Competitive nomination-based fellowship for top final-year PhD candidates (1–2 per department).
- ACCESS Discover Project Award** 2026–2027
 - Awarded 150,000 computing credits (~\$10K–15K equivalent) in national cyberinfrastructure resources to support scientific discovery.

Teaching Experience

Purdue University
Teaching Assistant - CS587: Foundations of Deep Learning

West Lafayette, IN
Spring 2024

- Co-designed and graded course assignments and exam questions involving equivariant models, optimizing deep neural networks, generative models, and optimization layers.

Purdue University

West Lafayette, IN

Teaching Assistant - CS373: Data Mining and Machine Learning

Fall 2022, Spring 2023

- Co-designed course assignments and exam questions involving fundamentals of machine learning, perceptron, SVM, learning theory, and data privacy.

Purdue University

West Lafayette, IN

Teaching Assistant - CS334: Fundamentals of Computer Graphics

Fall 2021, Fall 2023

- Co-designed course assignments and exam questions involving GPU programming, ray tracing, texture mapping, and procedural modeling.

United International University

Dhaka, Bangladesh.

Lecturer

2019 - 2021

- Redesigned and instructed courses, including Data Structures and Algorithms, Simulation and Modeling, Intro to Computer Graphics, and Computer Networking.

Services

Professional Service

Conference Reviewer:

- Neural Information Processing Systems
- International Conference on Machine Learning
- International Conference on Learning Representations
- Computer Vision and Pattern Recognition,

Journal Reviewer:

- IET Computer Vision
- IEEE Transactions on Neural Networks and Learning Systems

Research with Undergraduates

Michael Cheng

Undergraduate, Purdue, Fall 2024 – Fall 2025. Conducting undergraduate research under the supervision of **Professor Raymond A. Yeh**. As a mentor, I receive weekly updates, provide feedback, set project directions, and offer hands-on guidance in project development and implementation.

Research topic: Equivariant machine learning for computer vision.

Changxiang Gao

Undergraduate, Purdue, Summer 2024 – Fall 2024. Conducted undergraduate research under the supervision of **Professor Raymond A. Yeh**. As a mentor, I received weekly updates, defined project goals, and provided hands-on guidance for implementation.

Research topic: Any-resolution image classification.

Anugu Arun Reddy

Visiting Scholar, Purdue, Summer 2024 under the supervision of **Professor Aniket Bera**. As a mentor, I defined the project scope, received weekly updates, and guided the implementation process.

Research topic: Human motion generation.